

Section B

Task 1

Q. Create a database named foodapp_db and a table named restaurants with columns: restaurant_id (INT, AUTO_INCREMENT, PRIMARY KEY), name (VARCHAR, NOT NULL), city (VARCHAR), cuisine_type (VARCHAR), and rating (DECIMAL(3,1)).

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
4 CREATE DATABASE foodapp_db;
5
6 CREATE TABLE restaurants(
7     restaurant_id INTEGER PRIMARY KEY AUTO_INCREMENT,
8     name VARCHAR(100) NOT NULL,
9     city VARCHAR(100),
10    cuisine_type VARCHAR(100),
11    rating DECIMAL(3,1));
```

The Result Grid shows the execution of the SQL statements:

#	Time	Action	Message	Duration / Fetch
27	16:57:59	USE foodapp_db	0 row(s) affected	0.015 sec
28	17:00:41	CREATE TABLE restaurants(restaurant_id INTEGER PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100) NOT NULL, city VARCHAR(100), cuisine_type VARCHAR(100), rating DECIMAL(3,1));	0 row(s) affected	0.079 sec
29	17:07:08	INSERT INTO restaurants(name,city,cuisine_type,rating)VALUES ("Krishna Resort","Diu","Non-Veg",5), ("Radhika Resort","Diu","Non-Veg",5), ("Maheshwari","Morbi","Fast Food",4), ("Ajmeri","Rajkot","Punjabi Dish",4), ("Samrat Resort","Surat","Gujrati Dish",3), ("Rasal Resort","Vadodara","Chinease",4);	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0	0.000 sec
30	17:08:26	SELECT * FROM restaurants LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec

Q. Insert at least 6 restaurant records spanning at least 3 different cities.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
12 INSERT INTO restaurants(name,city,cuisine_type,rating)VALUES
13 ("Krishna Resort","Diu","Non-Veg",5),
14 ("Radhika Resort","Diu","Non-Veg",5),
15 ("Maheshwari","Morbi","Fast Food",4),
16 ("Ajmeri","Rajkot","Punjabi Dish",4),
17 ("Samrat Resort","Surat","Gujrati Dish",3),
18 ("Rasal Resort","Vadodara","Chinease",4);
```

The Result Grid shows the execution of the SQL statements:

#	Time	Action	Message	Duration / Fetch
27	16:57:59	USE foodapp_db	0 row(s) affected	0.015 sec
28	17:00:41	CREATE TABLE restaurants(restaurant_id INTEGER PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100) NOT NULL, city VARCHAR(100), cuisine_type VARCHAR(100), rating DECIMAL(3,1));	0 row(s) affected	0.079 sec
29	17:07:08	INSERT INTO restaurants(name,city,cuisine_type,rating)VALUES ("Krishna Resort","Diu","Non-Veg",5), ("Radhika Resort","Diu","Non-Veg",5), ("Maheshwari","Morbi","Fast Food",4), ("Ajmeri","Rajkot","Punjabi Dish",4), ("Samrat Resort","Surat","Gujrati Dish",3), ("Rasal Resort","Vadodara","Chinease",4);	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0	0.000 sec
30	17:08:26	SELECT * FROM restaurants LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec

Q. Update the rating of one restaurant and delete one restaurant record using a WHERE condition.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
17 ("Samrat Resort", "Surat", "Gujrati Dish", 3),
18 ("Rasal Resort", "Vadodara", "Chinease", 4);
19
20 SELECT * FROM restaurants;
21
22 UPDATE restaurants SET rating=3.5 WHERE restaurant_id=3;
23 DELETE FROM restaurants WHERE restaurant_id=5;
```

The Result Grid shows the output of the queries:

restaurant_id	name	city	cuisine_type	rating
1	Krishna Resort	Diu	Non-Veg	5.0
2	Radhika Resort	Diu	Non-Veg	5.0
3	Mareshwari	Morbi	Fast Food	3.5
4	Almeri	Rajkot	Punjabi Dish	4.0
6	Rasal Resort	Vadodara	Chinease	4.0

The Output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
31	17:19:49	UPDATE restaurants SET rating=3.5 WHERE restaurant_id=3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
32	17:19:54	SELECT * FROM restaurants LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec
33	17:21:18	DELETE FROM restaurants WHERE restaurant_id=5	1 row(s) affected	0.000 sec
34	17:21:21	SELECT * FROM restaurants LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Q. Write a SELECT query that retrieves all restaurants sorted by rating in descending order.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
19
20 SELECT * FROM restaurants;
21
22 UPDATE restaurants SET rating=3.5 WHERE restaurant_id=3;
23 DELETE FROM restaurants WHERE restaurant_id=5;
24
25 SELECT * FROM restaurants ORDER BY rating DESC;
```

The Result Grid shows the output of the queries:

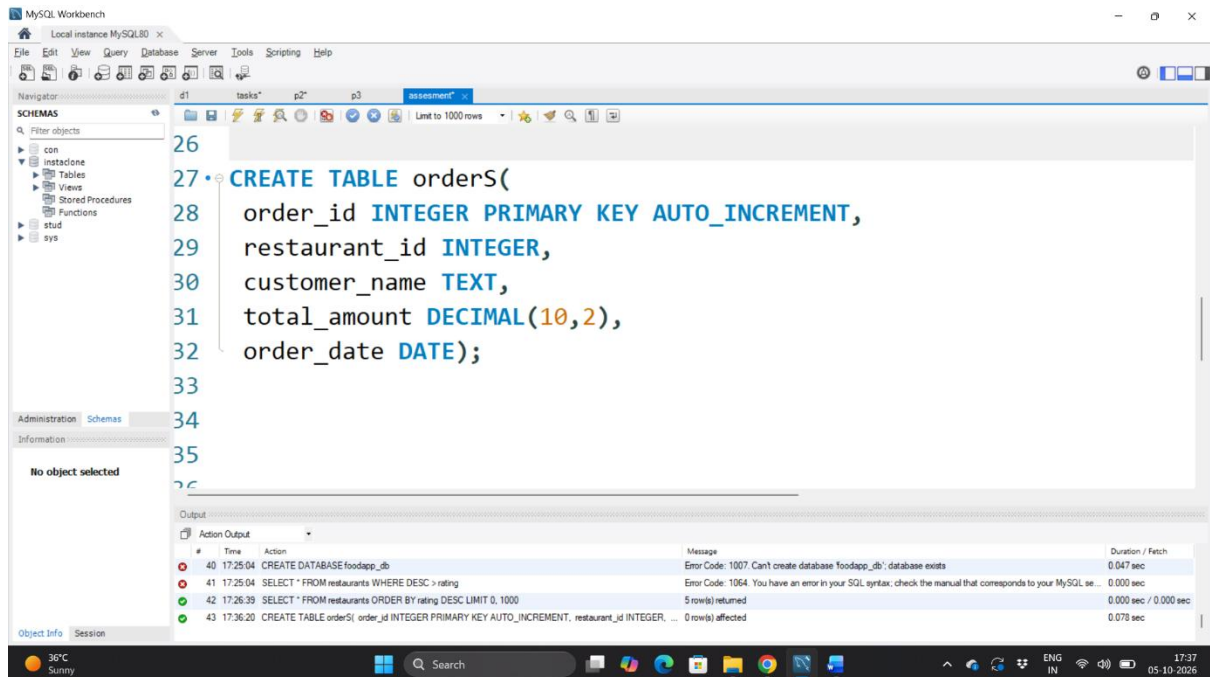
restaurant_id	name	city	cuisine_type	rating
1	Krishna Resort	Diu	Non-Veg	5.0
2	Radhika Resort	Diu	Non-Veg	5.0
4	Almeri	Rajkot	Punjabi Dish	4.0
6	Rasal Resort	Vadodara	Chinease	4.0
3	Mareshwari	Morbi	Fast Food	3.5

The Output pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
39	17:24:58	SELECT * FROM restaurants WHERE DESC rating	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
40	17:25:04	CREATE DATABASE foodapp_db	Error Code: 1007. Can't create database 'foodapp_db'; database exists	0.047 sec
41	17:25:04	SELECT * FROM restaurants WHERE DESC > rating	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
42	17:26:39	SELECT * FROM restaurants ORDER BY rating DESC LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Task 2

Q. Create an orders table with columns: order_id, restaurant_id, customer_name, total_amount, and order_date. Insert at least 12 orders distributed across at least 4 restaurants.

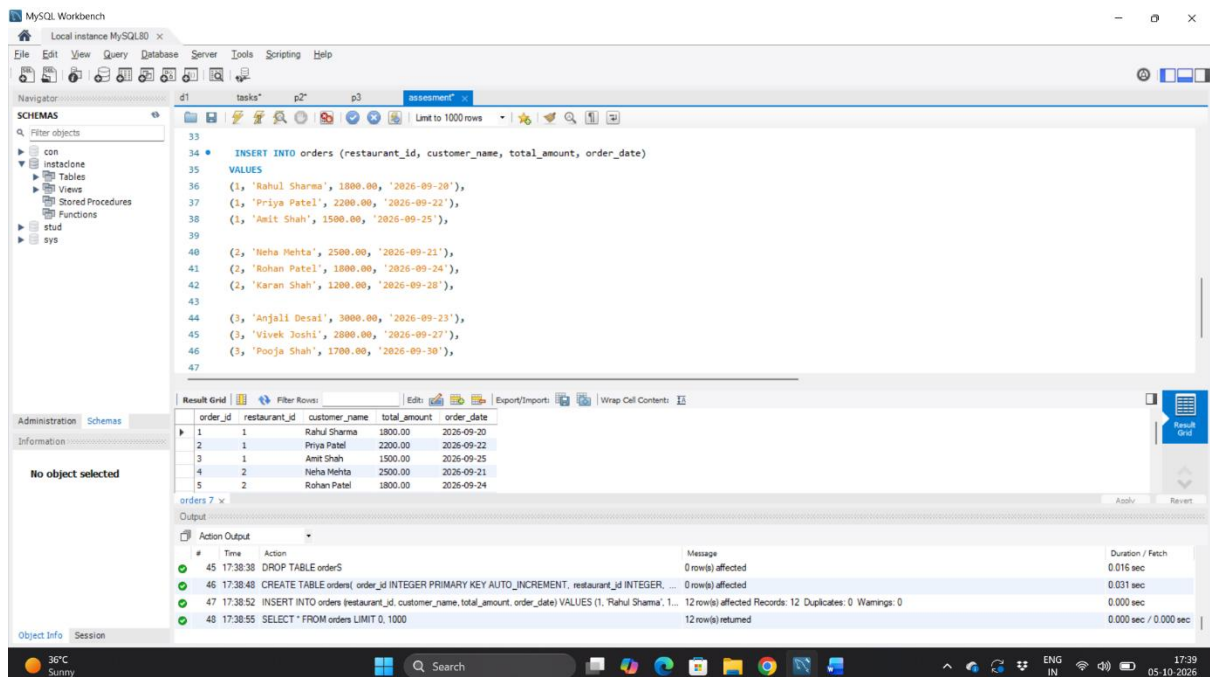


The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
26
27 CREATE TABLE orders(
28     order_id INTEGER PRIMARY KEY AUTO_INCREMENT,
29     restaurant_id INTEGER,
30     customer_name TEXT,
31     total_amount DECIMAL(10,2),
32     order_date DATE);
33
34
35
```

The Output tab shows the execution results:

#	Time	Action	Message	Duration / Fetch
40	17:25:04	CREATE DATABASE foodapp_db	Error Code: 1007. Can't create database 'foodapp_db': database exists	0.047 sec
41	17:25:04	SELECT * FROM restaurants WHERE DESC > rating	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
42	17:26:39	SELECT * FROM restaurants ORDER BY rating DESC LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
43	17:36:20	CREATE TABLE orders(order_id INTEGER PRIMARY KEY AUTO_INCREMENT, restaurant_id INTEGER, ...	0 row(s) affected	0.078 sec



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
33
34 INSERT INTO orders (restaurant_id, customer_name, total_amount, order_date)
35 VALUES
36 (1, 'Rahul Sharma', 1800.00, '2026-09-20'),
37 (1, 'Priya Patel', 2200.00, '2026-09-22'),
38 (1, 'Amit Shah', 1500.00, '2026-09-25'),
39
40 (2, 'Neha Mehta', 2500.00, '2026-09-21'),
41 (2, 'Rohan Patel', 1800.00, '2026-09-24'),
42 (2, 'Karan Shah', 1200.00, '2026-09-28'),
43
44 (3, 'Anjali Desai', 3000.00, '2026-09-23'),
45 (3, 'Vivek Joshi', 2800.00, '2026-09-27'),
46 (3, 'Pooja Shah', 1700.00, '2026-09-30'),
47
```

The Result Grid shows the following data:

order_id	restaurant_id	customer_name	total_amount	order_date
1	1	Rahul Sharma	1800.00	2026-09-20
2	1	Priya Patel	2200.00	2026-09-22
3	1	Amit Shah	1500.00	2026-09-25
4	2	Neha Mehta	2500.00	2026-09-21
5	2	Rohan Patel	1800.00	2026-09-24

The Output tab shows the execution results:

#	Time	Action	Message	Duration / Fetch
45	17:38:38	DROP TABLE orders	0 row(s) affected	0.016 sec
46	17:38:48	CREATE TABLE orders(order_id INTEGER PRIMARY KEY AUTO_INCREMENT, restaurant_id INTEGER, ...	0 row(s) affected	0.031 sec
47	17:38:52	INSERT INTO orders (restaurant_id, customer_name, total_amount, order_date) VALUES (1, 'Rahul Sharma', 1...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0	0.000 sec
48	17:38:55	SELECT * FROM orders LIMIT 0, 1000	12 row(s) returned	0.000 sec / 0.000 sec

Q. Write a query to show total revenue and average order value per restaurant.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

49 (4, 'Meera Patel', 2100.00, '2026-10-01'),
50 (4, 'Dev Shah', 1900.00, '2026-10-03');
51 • SELECT * FROM orders;
52
53 • SELECT restaurant_id,
54         SUM(total_amount) AS total_revenue,
55         AVG(total_amount) AS average_order_value
56 FROM orders
57 GROUP BY restaurant_id;

```

The Result Grid shows the following data:

restaurant_id	total_revenue	average_order_value
1	5000.00	1833.333333
2	5000.00	1833.333333
3	7500.00	2500.000000
4	5400.00	1800.000000

The Action Output pane shows the execution of the query, including the creation of the 'orders' table and the execution of the SELECT query.

Q. Write a query using HAVING to list only restaurants whose total revenue exceeds Rs 5,000

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

58
59 • SELECT restaurant_id,
60         SUM(total_amount) AS total_revenue
61 FROM orders
62 GROUP BY restaurant_id
63 HAVING SUM(total_amount) > 5000;

```

The Result Grid shows the following data:

restaurant_id	total_revenue
1	5000.00
3	7500.00
4	5400.00

The Action Output pane shows the execution of the query, including the execution of the SELECT query with the HAVING clause.

Q. Write a query to retrieve the 5 most recent orders using ORDER BY and LIMIT.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
53 HAVING SUM(total_amount) > 5000;
54
55 SELECT *
56 FROM orders
57 ORDER BY order_date DESC
58 LIMIT 5;
```

The Results grid displays the following data:

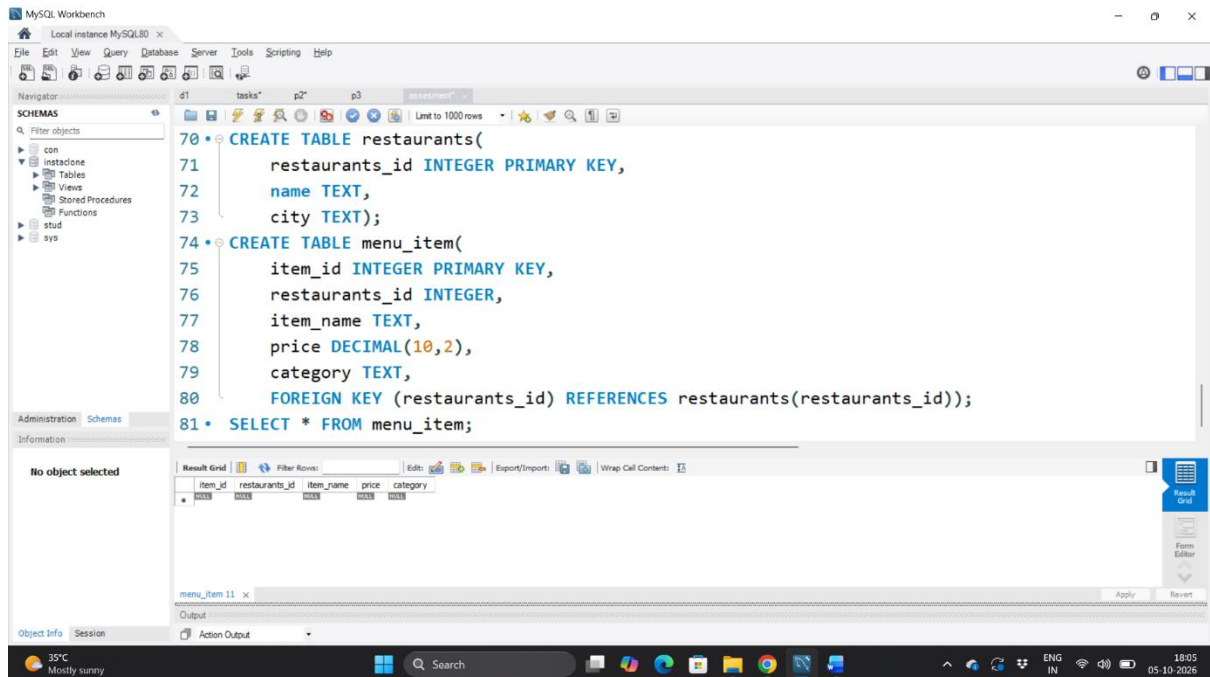
order_id	restaurant_id	customer_name	total_amount	order_date
12	4	Dev Shah	1900.00	2026-10-03
11	4	Meera Patel	2100.00	2026-10-01
9	3	Pooja Shah	1700.00	2026-09-30
6	2	Karan Shah	1200.00	2026-09-28
8	3	Vivek Joshi	2800.00	2026-09-27

The Output tab shows the execution log:

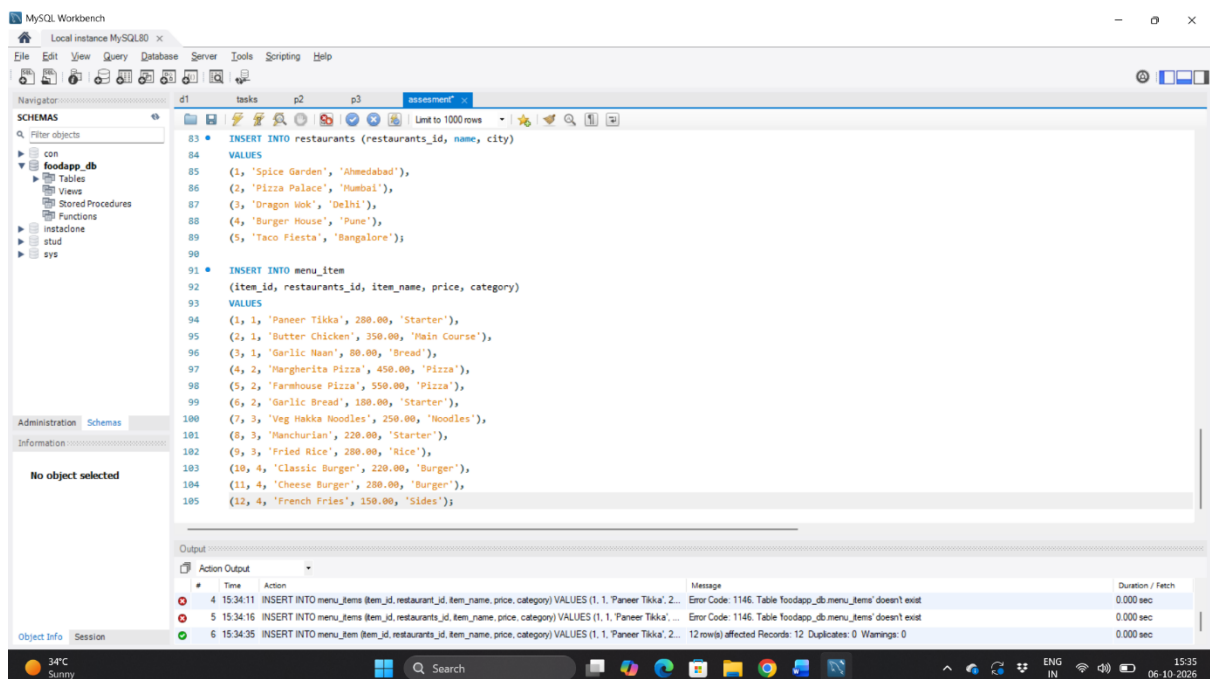
#	Time	Action	Message	Duration / Fetch
48	17:38:55	SELECT * FROM orders LIMIT 0, 1000	12 row(s) returned	0.000 sec / 0.000 sec
49	17:43:15	SELECT restaurant_id, SUM(total_amount) AS total_revenue, AVG(total_amount) AS average_order...	4 row(s) returned	0.000 sec / 0.000 sec
50	17:46:05	SELECT restaurant_id, SUM(total_amount) AS total_revenue FROM orders GROUP BY restaurant_id HAV...	4 row(s) returned	0.000 sec / 0.000 sec
51	17:49:40	SELECT * FROM orders ORDER BY order_date DESC LIMIT 5	5 row(s) returned	0.000 sec / 0.000 sec

Task 3

Q. Create two tables: restaurants (restaurant_id, name, city) and menu_items (item_id, restaurant_id, item_name, price, category). Link them with a FOREIGN KEY on restaurant_id.



Q. Insert at least 5 restaurants and 12 menu items; at least one restaurant must have no menu items.



MySQL Workbench

Local instance MySQL80

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Navigator

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SCHEMAS

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Administration Schemas

Information

No object selected

Limit to 1000 rows

Query Editor

```

L03 (10, 4, 'Classic Burger', 220.00, 'Burger'),
L04 (11, 4, 'Cheese Burger', 280.00, 'Burger'),
L05 (12, 4, 'French Fries', 150.00, 'Sides');
L06
L07 • SELECT * FROM menu_item;

```

Result Grid

Item_id	restaurants_id	Item_name	price	category
1	1	Panzer Tikka	280.00	Starter
2	1	Butter Chicken	350.00	Main Course
3	1	Garlic Naan	80.00	Bread
4	2	Margherita Pizza	450.00	Pizza
5	2	Farmhouse Pizza	550.00	Pizza
6	2	Garlic Bread	180.00	Starter
7	3	Veg Hakka Noodles	250.00	Noodles
8	3	Manchurian	220.00	Starter
9	3	Fried Rice	280.00	Rice
10	4	Classic Burger	220.00	Burger
11	4	Cheese Burger	280.00	Burger
12	4	French Fries	150.00	Sides

menu_item 1 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	15:34:16	INSERT INTO menu_items (item_id, restaurants_id, item_name, price, category) VALUES (1, 1, 'Panzer Tikka', ...	Error Code: 1146. Table 'foodapp_db.menu_items' doesn't exist	0.000 sec
6	15:34:35	INSERT INTO menu_items (item_id, restaurants_id, item_name, price, category) VALUES (1, 1, 'Panzer Tikka', ...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0	0.000 sec
7	15:35:34	SELECT * FROM menu_item LIMIT 0, 1000	12 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

34°C Sunny

15:35 06-10-2026

Q. Write an INNER JOIN query listing restaurant name, item name, and price for all available items.

MySQL Workbench

Local instance MySQL80

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Navigator

Filter objects

SCHEMAS

- con
- foodapp_db
 - Tables
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- instadone
- stud
- sys

Administration Schemas

Information

No object selected

Limit to 1000 rows

Query Editor

```

L06
L07 • SELECT * FROM menu_item;
L08
L09 • SELECT r.name AS restaurant_name, m.item_name, m.price
L10 FROM restaurants r INNER JOIN menu_item AS m
L11 ON r.restaurants_id = m.restaurants_id;

```

Result Grid

restaurant_name	Item_name	price
Spice Garden	Panzer Tikka	280.00
Spice Garden	Butter Chicken	350.00
Spice Garden	Garlic Naan	80.00
Pizza Palace	Margherita Pizza	450.00
Pizza Palace	Farmhouse Pizza	550.00
Pizza Palace	Garlic Bread	180.00
Dragon Wok	Veg Hakka Noodles	250.00
Dragon Wok	Manchurian	220.00
Dragon Wok	Fried Rice	280.00
Burger House	Classic Burger	220.00
Burger House	Cheese Burger	280.00
Burger House	French Fries	150.00

Result 2 x

Output

Action Output

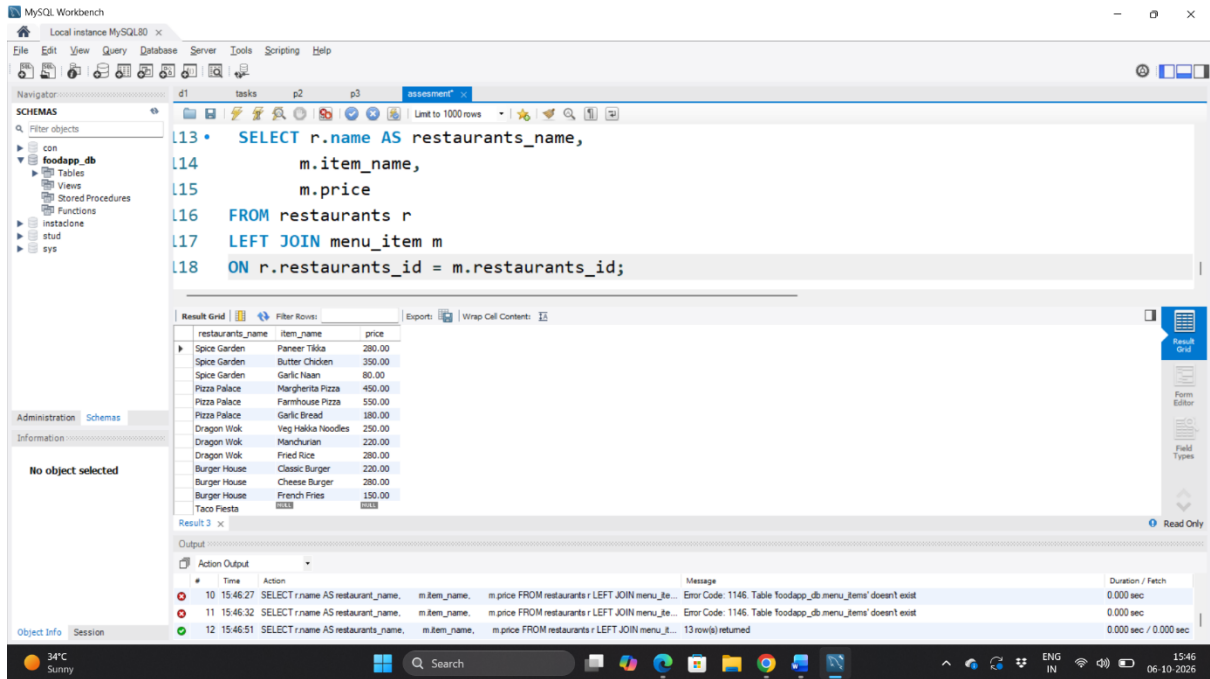
#	Time	Action	Message	Duration / Fetch
7	15:35:34	SELECT * FROM menu_item LIMIT 0, 1000	12 row(s) returned	0.016 sec / 0.000 sec
8	15:43:36	SELECT r.name AS restaurant_name, m.item_name, m.price FROM restaurants r INNER JOIN menu_item AS m ON ...	Error Code: 1054. Unknown column 'm.item' in field list	0.000 sec
9	15:44:01	SELECT r.name AS restaurant_name, m.item_name, m.price FROM restaurants r INNER JOIN menu_item AS ...	12 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

34°C Sunny

15:44 06-10-2026

Q. Write a LEFT JOIN query so that restaurants with no items also appear in results, with NULL values in item columns.



The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```

113 • SELECT r.name AS restaurants_name,
114       m.item_name,
115       m.price
116 FROM restaurants r
117 LEFT JOIN menu_item m
118 ON r.restaurants_id = m.restaurants_id;

```

The Result Grid displays the following data:

restaurants_name	item_name	price
Spice Garden	Panzer Tella	280.00
Spice Garden	Butter Chicken	350.00
Spice Garden	Garlic Naan	80.00
Pizza Palace	Margherita Pizza	450.00
Pizza Palace	Farmhouse Pizza	550.00
Pizza Palace	Garlic Bread	180.00
Dragon Wok	Veg Noodle Noodles	250.00
Dragon Wok	Manchurian	220.00
Dragon Wok	Fried Rice	280.00
Burger House	Classic Burger	220.00
Burger House	Cheese Burger	280.00
Burger House	French Fries	150.00
Taco Fiesta		

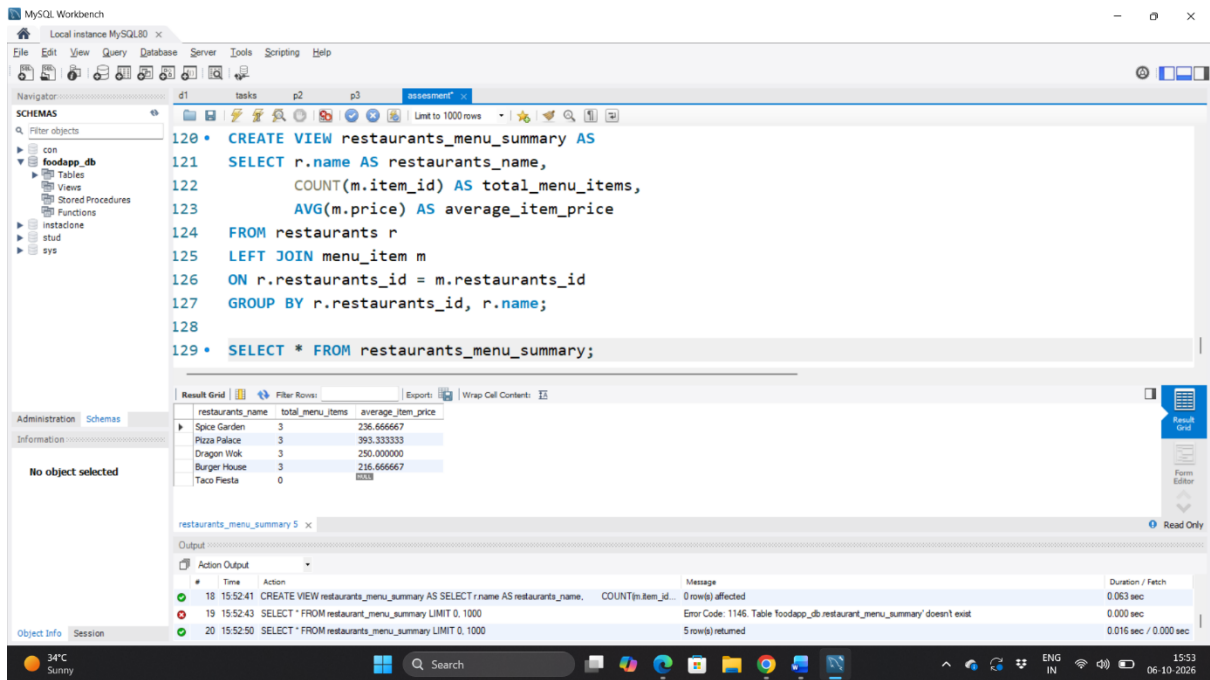
The Output pane shows the execution log with the following messages:

```

10 15:46:27 SELECT r.name AS restaurants_name, m.item_name, m.price FROM restaurants r LEFT JOIN menu_item m
11 15:46:32 SELECT r.name AS restaurants_name, m.item_name, m.price FROM restaurants r LEFT JOIN menu_item m
12 15:46:51 SELECT r.name AS restaurants_name, m.item_name, m.price FROM restaurants r LEFT JOIN menu_item m

```

Q. Create a VIEW named restaurant_menu_summary showing restaurant name, total number of menu items, and average item price grouped by restaurant.



The screenshot shows the MySQL Workbench interface. The query editor contains the following SQL code:

```

120 • CREATE VIEW restaurant_menu_summary AS
121 SELECT r.name AS restaurants_name,
122        COUNT(m.item_id) AS total_menu_items,
123        AVG(m.price) AS average_item_price
124 FROM restaurants r
125 LEFT JOIN menu_item m
126 ON r.restaurants_id = m.restaurants_id
127 GROUP BY r.restaurants_id, r.name;
128
129 • SELECT * FROM restaurant_menu_summary;

```

The Result Grid displays the following data:

restaurants_name	total_menu_items	average_item_price
Spice Garden	3	236.666667
Pizza Palace	3	393.333333
Dragon Wok	3	250.000000
Burger House	3	216.666667
Taco Fiesta	0	

The Output pane shows the execution log with the following messages:

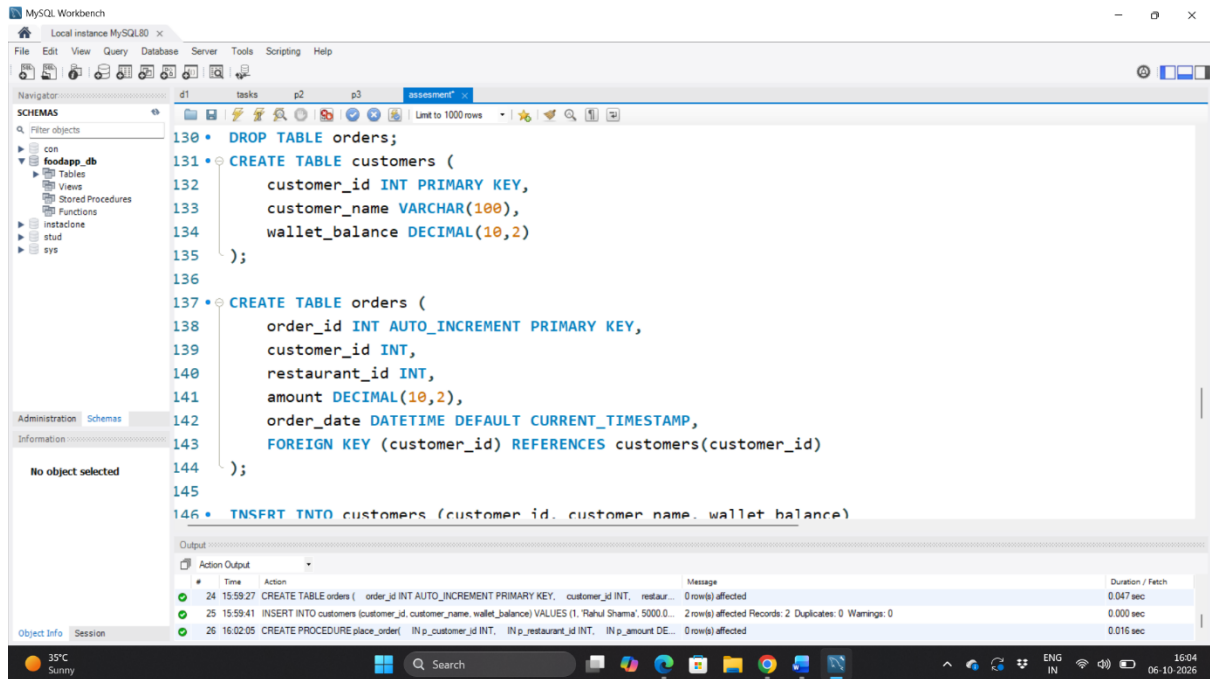
```

18 15:52:41 CREATE VIEW restaurant_menu_summary AS SELECT r.name AS restaurants_name, COUNT(m.item_id) AS total_menu_items,
19 15:52:43 SELECT * FROM restaurant_menu_summary LIMIT 0, 1000
20 15:52:50 SELECT * FROM restaurant_menu_summary LIMIT 0, 1000

```

Task 4

Q. Create a stored procedure named `place_order` that accepts `customer_id`, `restaurant_id`, and `amount` as parameters.



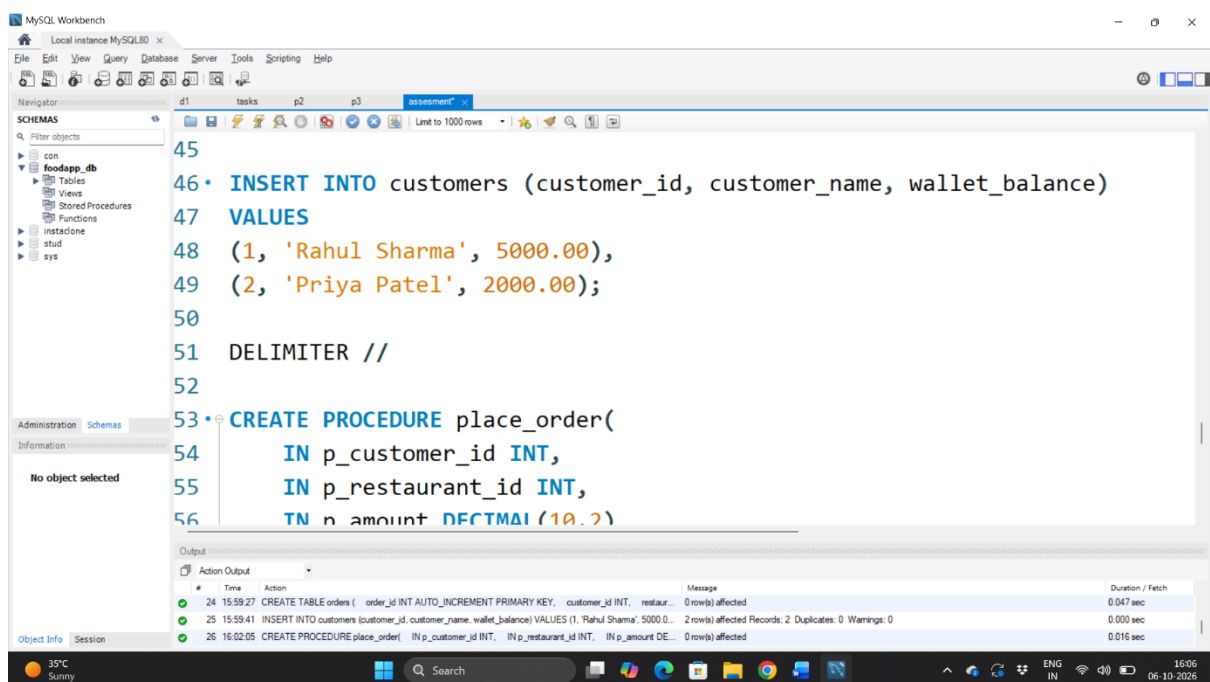
The screenshot shows the MySQL Workbench interface with the following SQL queries in the editor:

```
130 • DROP TABLE orders;
131 • CREATE TABLE customers (
132     customer_id INT PRIMARY KEY,
133     customer_name VARCHAR(100),
134     wallet_balance DECIMAL(10,2)
135 );
136
137 • CREATE TABLE orders (
138     order_id INT AUTO_INCREMENT PRIMARY KEY,
139     customer_id INT,
140     restaurant_id INT,
141     amount DECIMAL(10,2),
142     order_date DATETIME DEFAULT CURRENT_TIMESTAMP,
143     FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
144 );
145
146 • INSERT INTO customers (customer_id, customer_name, wallet_balance)
```

The Output window shows the following results:

#	Time	Action	Message	Duration / Fetch
24	15:59:27	CREATE TABLE orders (order_id INT AUTO_INCREMENT PRIMARY KEY, customer_id INT, restaur...	0 row(s) affected	0.047 sec
25	15:59:41	INSERT INTO customers (customer_id, customer_name, wallet_balance) VALUES (1, 'Rahul Sharma', 5000.00...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
26	16:02:05	CREATE PROCEDURE place_order(IN p_customer_id INT, IN p_restaurant_id INT, IN p_amount DE...	0 row(s) affected	0.016 sec

Q. Deduct the amount from the customer's `wallet_balance` in a `customers` table and insert a new record into the `orders` table.



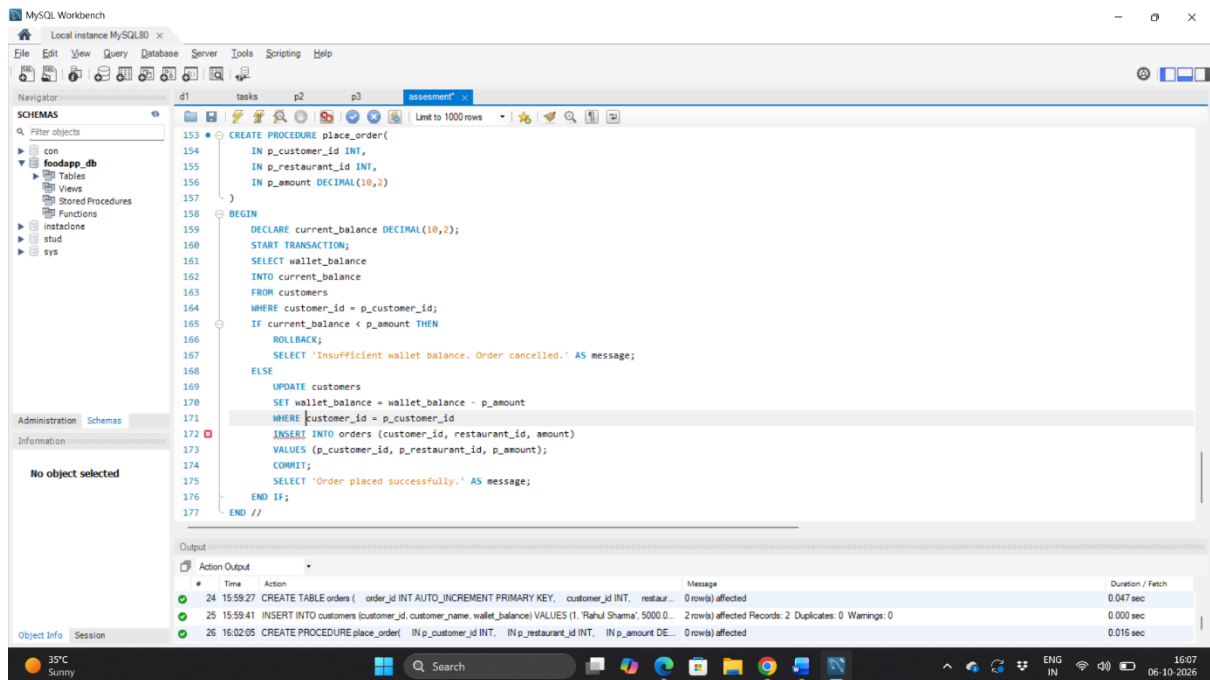
The screenshot shows the MySQL Workbench interface with the following SQL queries in the editor:

```
45
46 • INSERT INTO customers (customer_id, customer_name, wallet_balance)
47 VALUES
48 (1, 'Rahul Sharma', 5000.00),
49 (2, 'Priya Patel', 2000.00);
50
51 DELIMITER //
52
53 • CREATE PROCEDURE place_order(
54     IN p_customer_id INT,
55     IN p_restaurant_id INT,
56     IN p_amount DECIMAL(10,2)
```

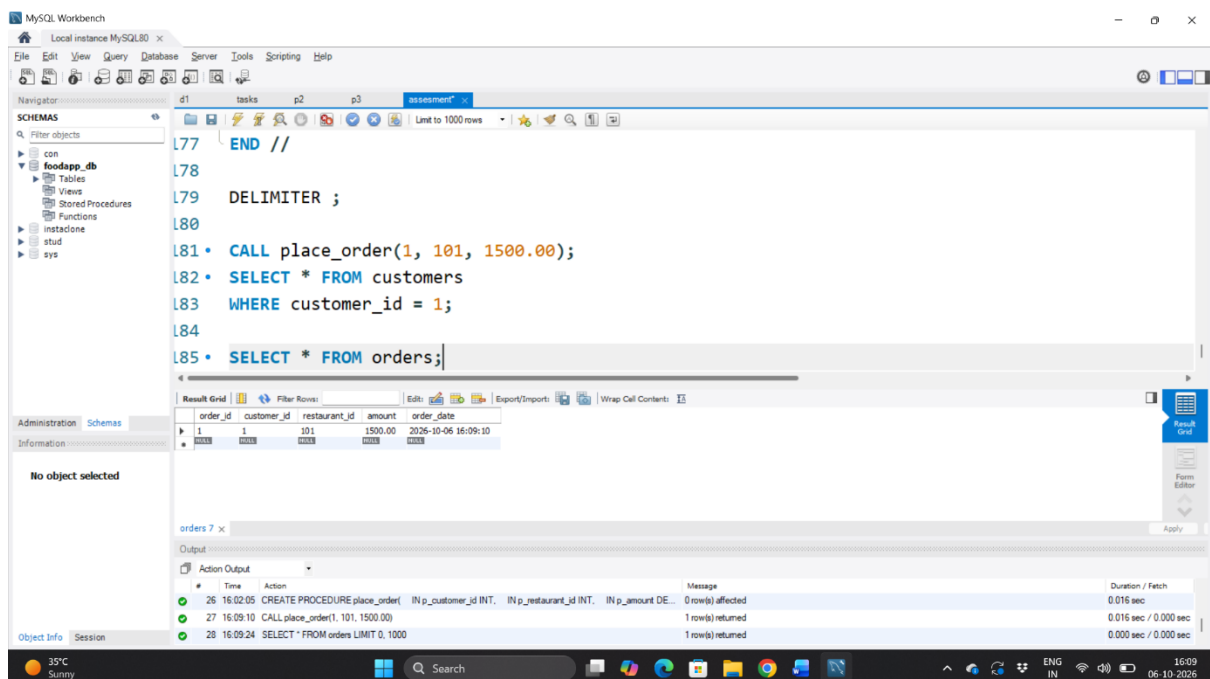
The Output window shows the following results:

#	Time	Action	Message	Duration / Fetch
24	15:59:27	CREATE TABLE orders (order_id INT AUTO_INCREMENT PRIMARY KEY, customer_id INT, restaur...	0 row(s) affected	0.047 sec
25	15:59:41	INSERT INTO customers (customer_id, customer_name, wallet_balance) VALUES (1, 'Rahul Sharma', 5000.00...	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
26	16:02:05	CREATE PROCEDURE place_order(IN p_customer_id INT, IN p_restaurant_id INT, IN p_amount DE...	0 row(s) affected	0.016 sec

Q. Use a transaction block: if the customer's wallet_balance is insufficient, ROLLBACK everything; otherwise COMMIT.



Q. Test with a valid order and an order exceeding wallet balance. Screenshot both outcomes from MySQL Workbench.



MySQL Workbench

Local instance MySQL80 x

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Navigator

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Filter objects

- con
- foodapp_db
 - Tables
 - Views
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- instadone
- stud
- sys

d1 tasks p2 p3 assessment*

Limit to 1000 rows

```
84
85 • SELECT * FROM orders;
86
87 • CALL place_order(2, 102, 3000.00);
88 • SELECT * FROM customers
89   WHERE customer_id = 2;
90
91 • SELECT * FROM orders;
```

Result Grid

order_id	customer_id	restaurant_id	amount	order_date
1	1	101	1500.00	2026-10-06 16:09:10

Information

No object selected

orders 9 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
28	16:09:24	SELECT * FROM orders LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
29	16:09:56	CALL place_order(2, 102, 3000.00)	1 row(s) returned	0.000 sec / 0.000 sec
30	16:10:10	SELECT * FROM orders LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

35°C Sunny

Search

ENG IN

16:10 06-10-2026